Drone Radiometric and BRDF Correction: What the Data Actually Support

Gordon, Goldhill and Ruby 2023 | Real mosaics, matched researcher polygons, shared validity gates

1. What information do the mosaics contain? Spatially validated ten-band imagery, terrain, smooth solar grids and scalar nadir view placeholders. The files do not identify the camera observations or blend weights contributing to each ortho pixel. Physical calibration and acquisition-date/timezone provenance remain unresolved.

2. Can any support the current Ross-Li fit? No. Every flight has rank one for three model parameters. Millions of cells and distinct floating-point solar values do not create independent viewing information. Thirty of thirty band fits are unsupported; no coefficients were forced or applied.

Flight	Valid source pixels	Rank / 3	Condition number	Effective dimensions	Supported / unsupported
Gordon	6,153,336	1	2.1e+24	1.0	0 / 10
Goldhill	7,068,376	1	2.2e+24	1.0	0 / 10
Ruby	2,022,614	1	3.6e+25	1.0	0 / 10

3. Which corrections improve consistency? Topographic DN sensitivity is mixed: variance falls in 8 of 12 sufficiently sampled flight/category cells, while median band CV falls in 6 and rises in 6. Gains reach 528x and coverage falls by 1.49-5.49%. This does not establish a blanket topographic correction benefit or physical accuracy.

4. How should drone and NEON differ? Share the equations and physical-unit contracts; require actual geometry and independent eligibility for every dataset. NEON directional-flightline observation geometry is a different input contract from these drone placeholders. Product revision and previous correction state must also be checked.

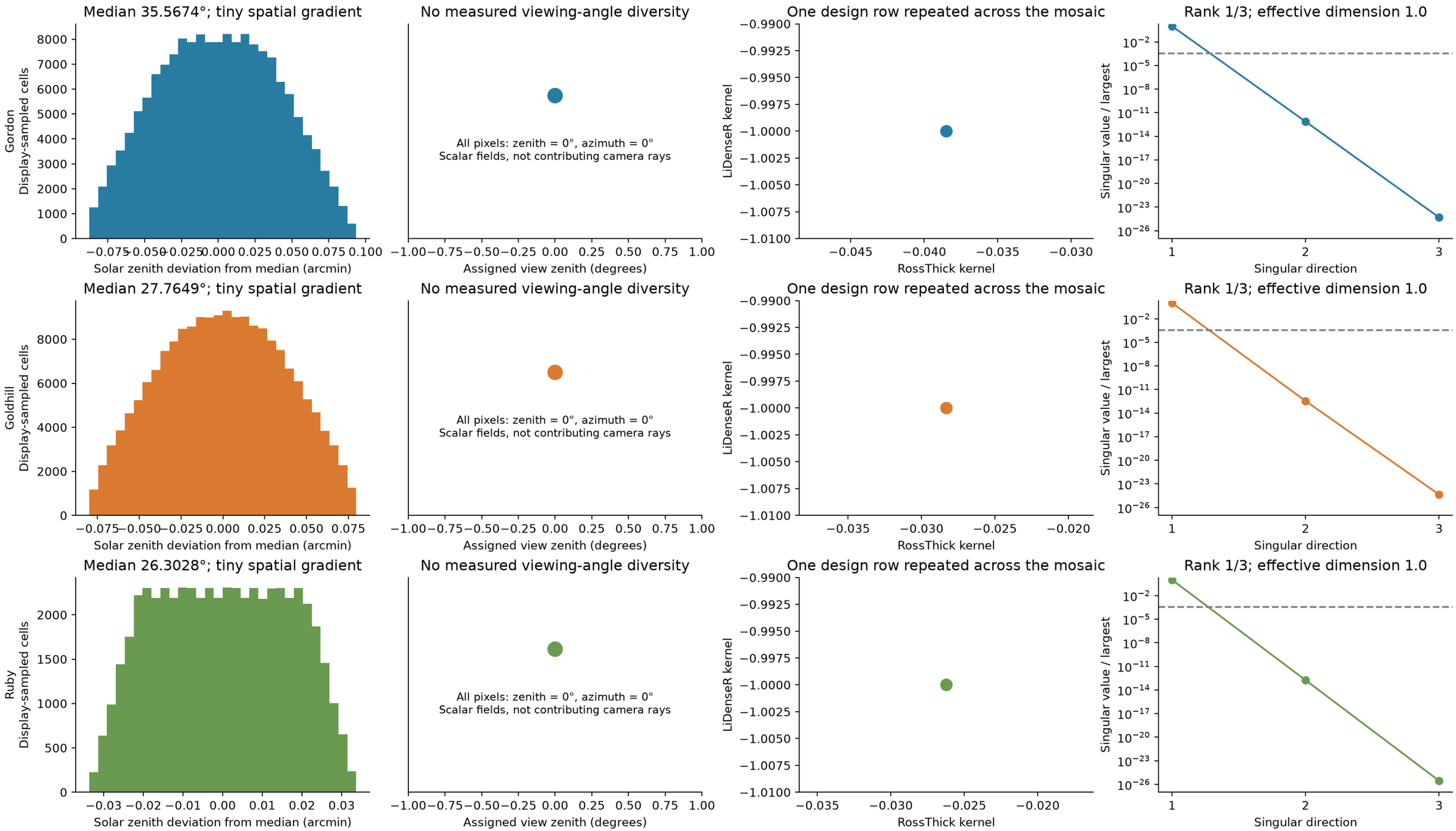
5. What should change? Preserve recovered DN as the temporary numerical baseline; omit unsupported BRDF; require calibration and validated illumination/terrain before promoting physical or topographic products. Use explicit stage states and scientific QA. Useful no-BRDF libraries are possible after those prerequisites, with a documented directional domain.

This is an isolated experiment, not a production repair. Raw-photo BRDF remains a future possibility if calibrated images, poses, repeated targets and stable illumination can be recovered. The available metadata CSVs contain filenames/times/software only; their August 15 timestamps conflict with August 14 notes. No globally exhaustive archive search or new empirical NEON cohort was performed.

Angular information: a rank-one design in all three flights

Full-scene rank diagnostics; small solar raster gradients are not recovered per-photo viewing angles

Figure 1. Millions of cells, one usable angular design direction per flight

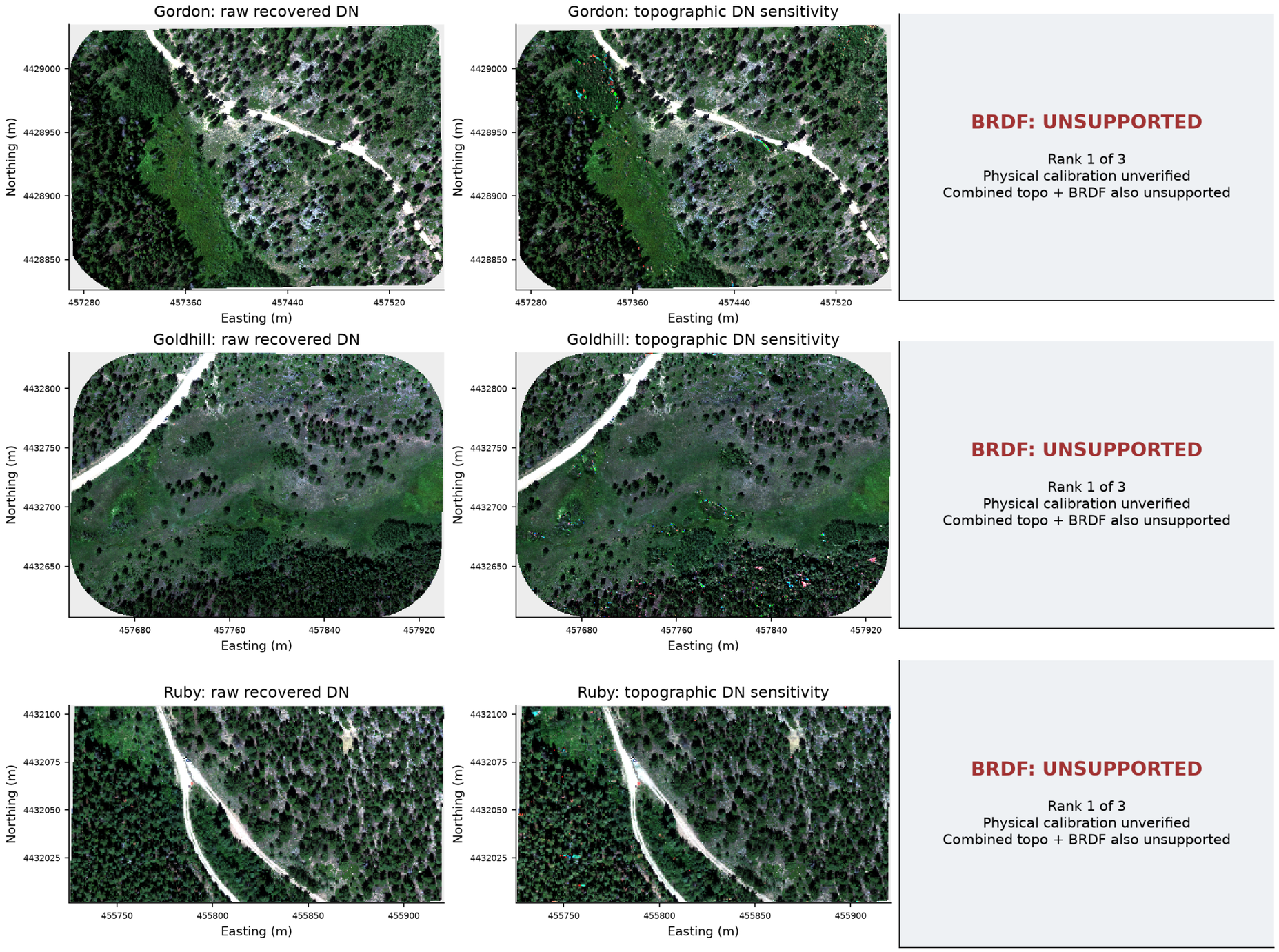


Rank and singular values use every source-valid spectral/angle location. Histograms are display-sampled only. Stable rank is an effective matrix dimension, not an independent-image sample count; that count is unknown. The unchanged condition limit is about 2,896; the observed condition numbers exceed 10^24.

Full-flight comparison: approved BRDF branches are unsupported

Same grid, RGB bands and raw-reference contrast per flight; topographic panels are numerical sensitivity only

Figure 2. Raw and topographic sensitivity; no valid BRDF output for any flight
Identical raw-reference RGB stretch within each flight; all values are source DN, not verified reflectance

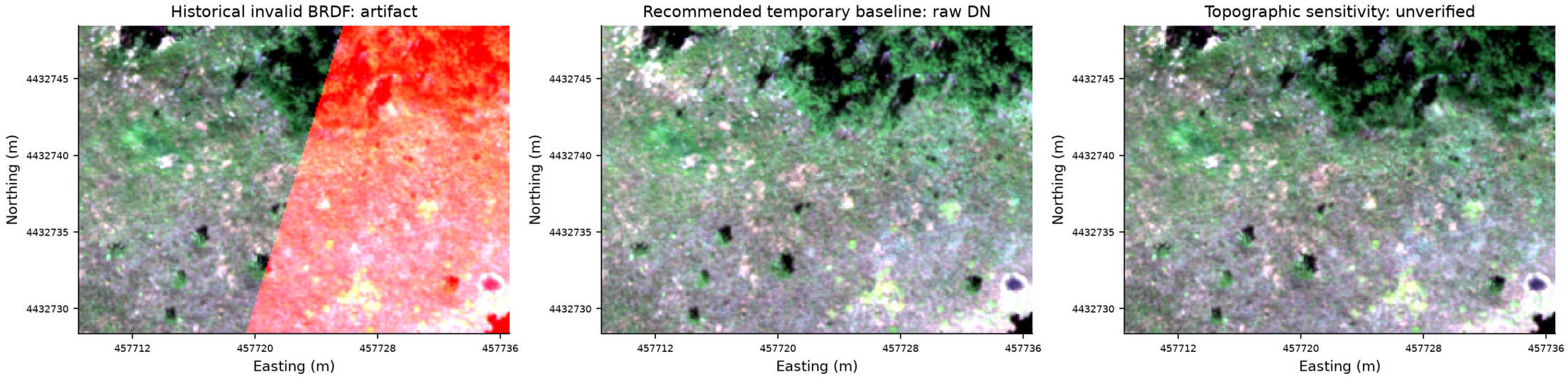


No valid BRDF or combined corrected raster was substituted. Topographic-only approval is also held: illumination, terrain and physical units are unverified. The shown SCS+C replay isolates numerical behavior with existing embedded geometry. Native-resolution matched regions are provided separately in Figure 2b and the Markdown report.

Goldhill: avoid the unsupported correction that creates the streak

Historical invalid model shown only as a diagnostic; current temporary baseline remains uncalibrated DN

Figure 3. Avoiding an unsupported correction avoids the artificial red boundary
Raw and topo diagnostic remain uncalibrated; neither is presented as validated physical reflectance



The earlier exact replay matched all 70,832,700 production values. Inverse normalization changed the red boundary metric by only 0.00000501%; the same historical factors still reach 11.7 million. The current repaired kernel has rank one, so it rejects this unsupported fit. Removing the artifact does not prove physical calibration.

Flight	Pixels lost to topo geometry	Maximum topo factor	Band-pixel factors above 10
Gordon	250,821 (4.08%)	528.41x	15,984
Goldhill	388,047 (5.49%)	356.55x	43,179
Ruby	30,171 (1.49%)	372.71x	2,665

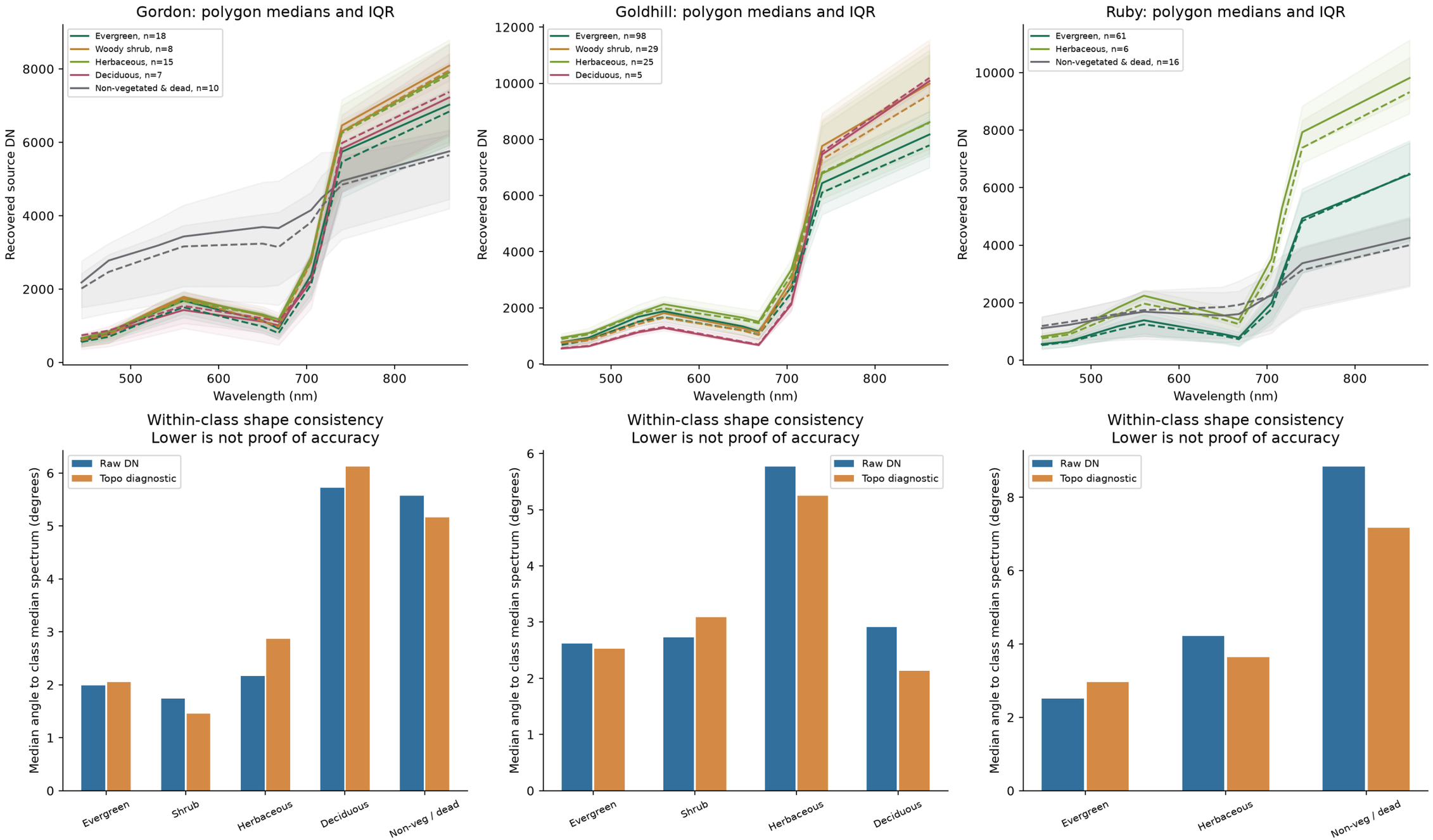
Large SCS+C factors and invalid ancillary masks require separate fit/denominator and terrain review. No gain cap was introduced. Topographic output is not approved merely because it avoids the red BRDF boundary. No additional fitted normalization was justified: invariant targets, calibrated overlaps and matched cross-flight polygons are unavailable.

Spectral consequences: matched polygons, mixed outcomes

One median spectrum per polygon; researcher labels preserved; lower variation alone is not accuracy

Figure 4. Topographic correction has mixed spectral effects

Solid = raw DN; dashed = topo sensitivity. One median per non-overlapping polygon component; minimum five polygons per category.

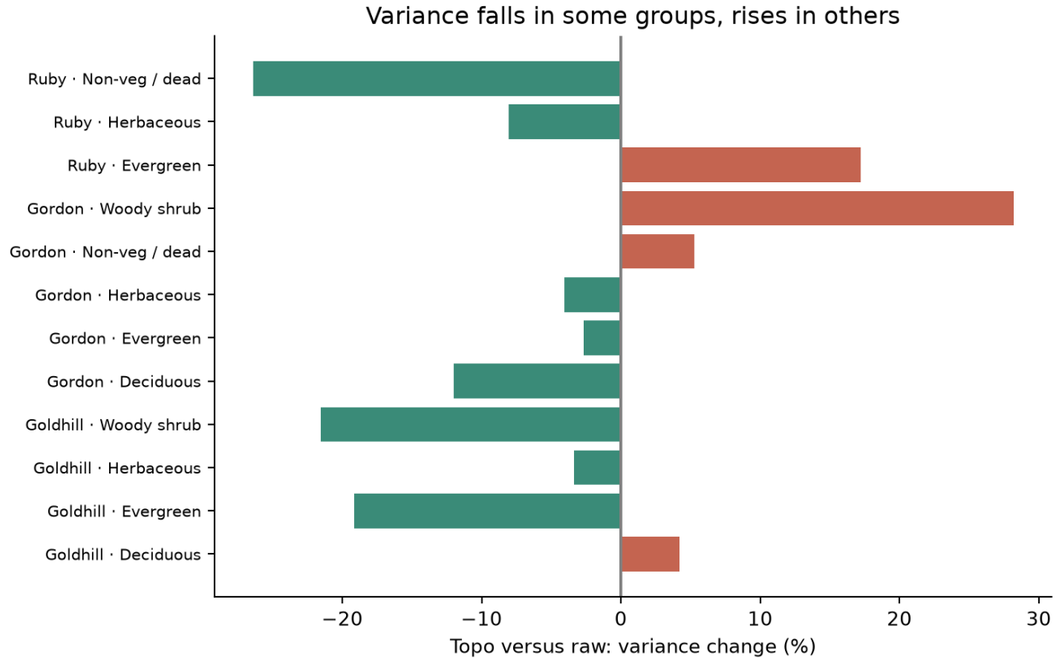


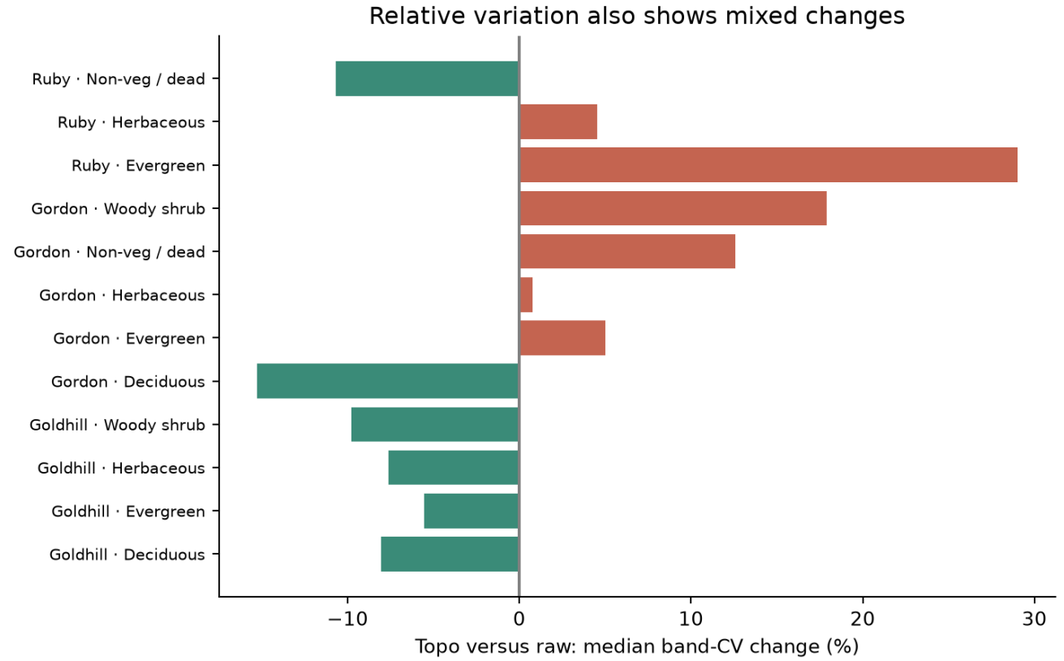
301 polygons and 198,309 matched polygon-pixel memberships were checked independently against geometry. Categories require at least five retained polygon units; Ruby shrubs are insufficient. Non-vegetated & dead is a cover comparator, not a biological PFT. Solid/dashed curves use the same raw/topographic-valid pixels; shaded ranges are polygon IQRs.

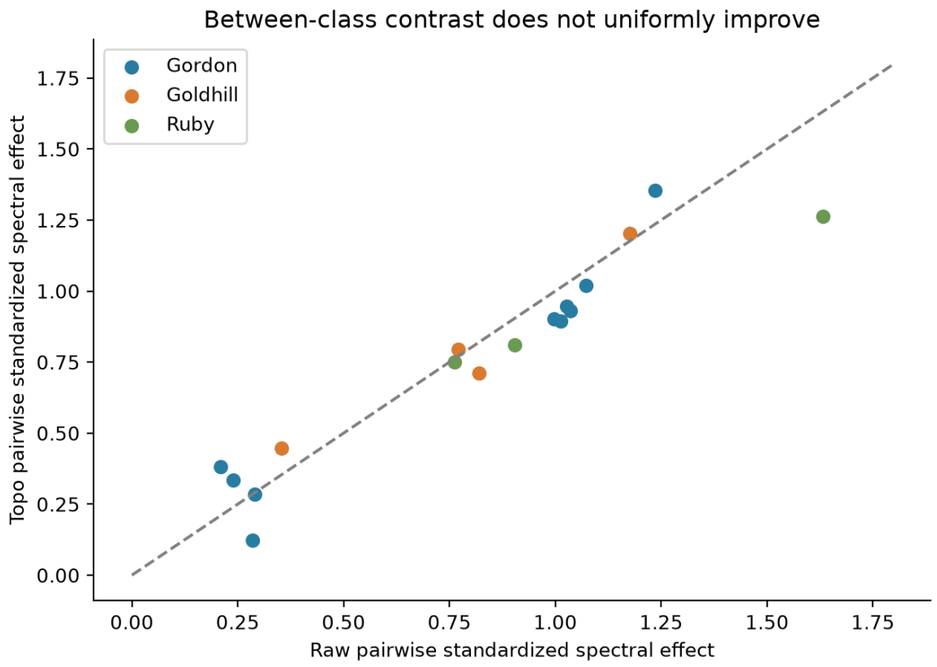
Correction performance: no uniform improvement

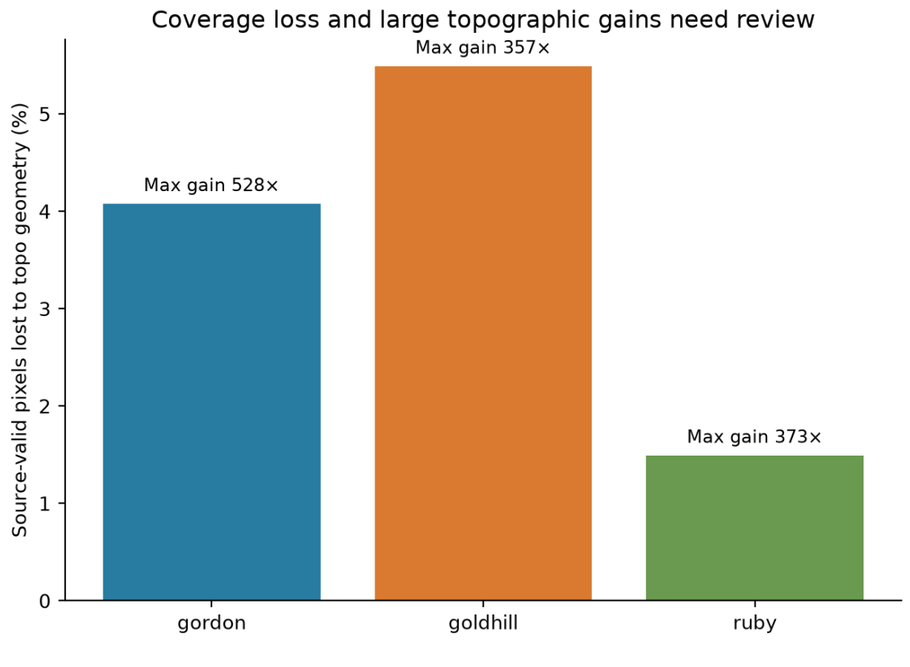
Numerical dispersion, spatial-blocked class contrasts and source-wide stability must be considered together

Figure 5. No blanket winner: numerical consistency is not physical validation
Matched polygons, identical pixels, spatial-blocked conditional comparisons; no pixel-random classification score.







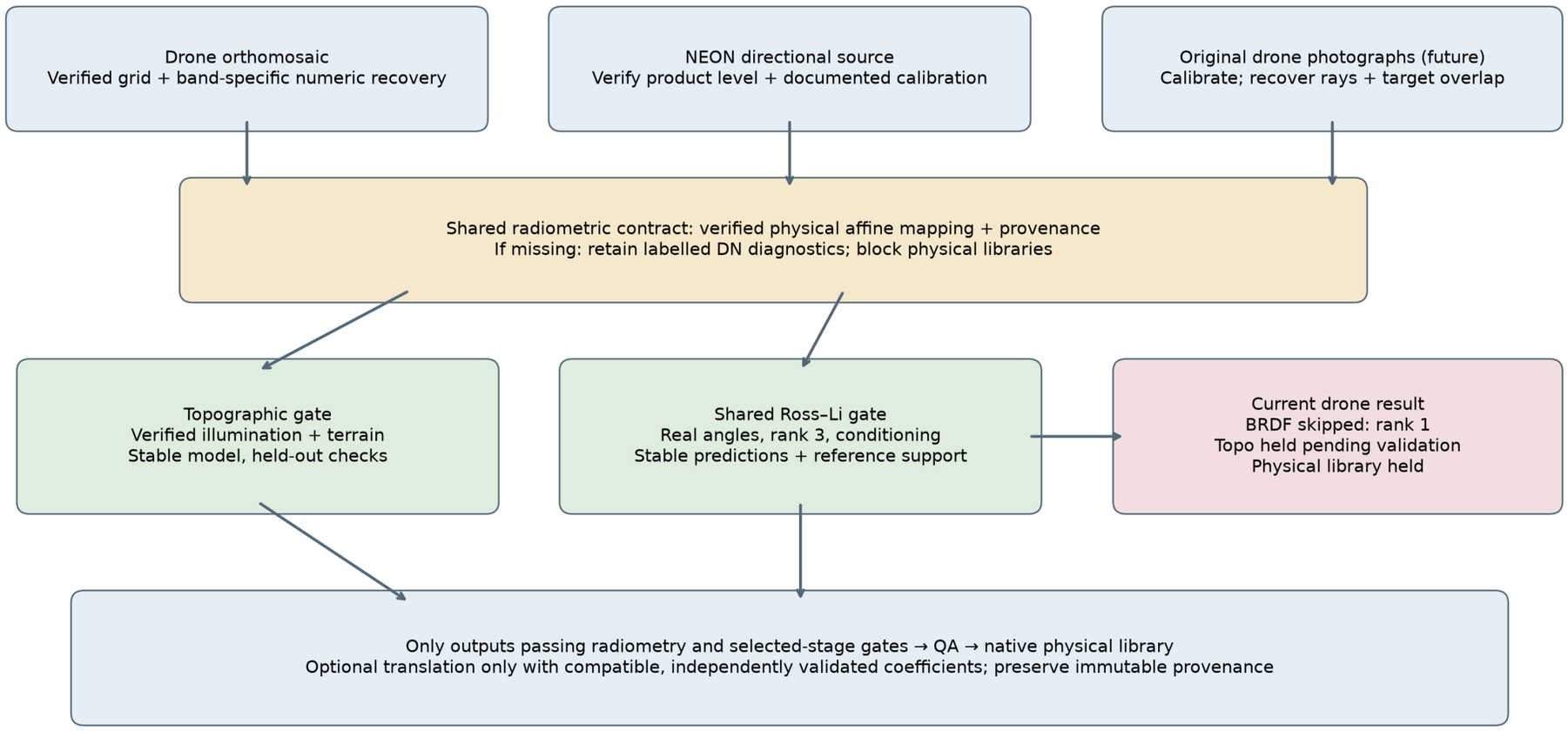


Ten-band class contrasts reuse the package separability API: 1,999 permutations within 100 m spatial blocks, one spectrum per identity, equal discrete band weights and Holm correction across 34 supported tests; 10 comparisons lack support. These are conditional observational associations, not prediction accuracy or tests of correction-caused improvement. No pixel-random split was used. Residual spatial dependence remains a limitation.

Proposed production workflow: share implementations, gate eligibility

Proposal only; preserve source grids and historical products, then approve calibration and correction policy before regeneration

Figure 6. Proposed policy: share the equations, require evidence at every stage



Acceptance: verified source calibration and per-band recovery provenance; authoritative date/timezone and terrain conventions; independent topographic stability and held-out checks; genuine angular provenance, rank, conditioning and supported reference predictions for BRDF; compatible physical-unit translation coefficients and native-library provenance.

Next data request: original SD1/SD2 photos, panel/DLS/exposure records, Metashape project or intrinsics/extrinsics, DSM and per-image target overlap. Test repeated targets before committing to pre-mosaic BRDF. Do not infer calibration from export range or camera-file count. Current libraries and dependent translated/bulk contributions remain historical/unapproved until a reviewed fresh run.

Reproduction and measured evidence: provenance/drone-brdf-strategy/2026-11-02/ (analyze.py, figures.py, report.py, metadata_audit.py, make_pdf.py; per-flight JSON and polygon/class tables). The Markdown report and separate production plan document assumptions, stage states, source hashes, original-archive search limits and narrow regression requirements.

Primary context: NEON radiance-to-reflectance ATBD revision B (data.neonscience.org/api/v0/documents/NEON.DOC.001288vB); NEON bidirectional reflectance tutorial (neonscience.org/resources/learning-hub/tutorials/neon-brdf-refl-h5-py); Agisoft MicaSense reflectance-calibration workflow (agisoft.freshdesk.com, solution 31000148780). Provider/vendor documentation does not validate these drone project settings.